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Severe and long-lasting hypotension occuring immediately after parathyroidectomy in hypertensive hemodialysis patients: a case series

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Secondary hyperparathyroidism of hemodialysis patients is associated with hypertension. Parathyroidectomy might cause a mild and gradual decrease in blood pressure. We describe a case series of severe and long-lasting hypotension starting immediately after parathyroidectomy, possibly related to resetting of intracellular calcium levels and attenuation of the vascular response to calcium.

Primary, secondary and tertiary forms of hyperparathyroidism are associated with hypertension.[1] Severe secondary hyperparathyroidism is an indication for corrective parathyroidectomy. Some studies have reported that parathyroidectomy induces a mild and gradual drop in blood pressure (5–10 mm Hg). Other studies showed no effect at all.[2–4] A recent work utilizing 24-h ambulatory blood pressure monitoring concluded that blood pressure increases after parathyroidectomy.[5] However, a substantial hypotensive effect has not yet been reported.

We present a case series of initially hypertensive end stage renal disease (ESRD) patients, who had an abrupt drop in blood pressure immediately after parathyroidectomy of hyperplastic parathyroid glands, leading to postoperative refractory hypotension persisting for months to years. Hypotension was documented at clinical measurements (peridialytic) and at repeated home blood pressure measurements.

Initially, patients underwent a workup for their hypotension to exclude any other cause (unrelated to the parathyroid operation) and to delineate possible mechanisms of parathyroidectomy-related hypotension. Patients had their supine and orthostatic blood pressure measured by a standard sphyngomanometer (Welch Allyn, Skaneateles Falls, NY, USA) after 5 min of lying down (two measurements) and within 2 min of standing (two measurements). Later, patients had their central (aortic) blood pressure and their pulse wave velocity (PWV) also measured (using SphygmoCor; Atcor Medical, Sydney, New South Wales, Australia).[6,7]

CASE I

A 45-year-old man had been on hemodialysis since 29 years of age. His average predialysis blood pressure was 140/90. He had been diagnosed with severe secondary hyperparathyroidism and thus underwent excision of four hyperplastic parathyroid glands 10 years ago. His immediate postoperative blood pressure was 60/40 mm Hg. His calcium and parathyroid hormone (PTH) values dropped abruptly (Table 1). In the last 10 years, since the operation, the patient has suffered from symptomatic hypotension, with an average blood pressure of 70/50 mm Hg. He underwent full endocrine evaluation (thyroid stimulating hormone, cortisol and adrenocorticotropic hormone) and cardiac function (echocardiogram), which were all normal. Results of his autonomic function studies suggested a hyperactive (compensatory) adrenergic response. An electromyography revealed only a mild peripheral neuropathy. In addition, his plasma renin activity and aldosterone levels were within normal range for upright position and normal sodium diet. A hemodynamic study (Impedance Cardiography; Bio Z ICG monitor, CardioDynamics, San Diego, CA, USA) revealed cardiac index of 3.8 l min^{-1} m^{-2} that increased at peak exercise to 8.3 ml min^{-1} m^{-2}. His systemic vascular resistance (SVR) at rest was low—586 dyn s cm^{-5} (normal 742–1378), and decreased even further during exercise—332 at peak exercise. His SVR index was similarly low at rest and during exercise. Trials of increased salt and coffee consumption, increase in sodium dialysate (145 mM), increase in dry weight (less ultrafiltration), increase in calcium dialysate (3 meq l^{-1}), decrease in dialysate temperature (35 °C) and administration of midodrine and fludrocortisone were all ineffective in normalizing his blood pressure. His central (aortic) blood pressure was 66/45 mm Hg. He had a PWV of 6 m s^{-1}, indicating very low arterial stiffness.

CASE II

A 46-year-old female suffered from resistant hypertension since 23 years of age, and was treated originally with clonidine, amlodipine and atenolol. Despite these interventions, she maintained a blood pressure of 180–200/100 mm Hg. Her blood pressure later stabilized after several years of hemodialysis (HD), and she remained only on an alpha blocker (doxazosin). She had long-standing hyperparathyroidism (PTH 7000 pg ml^{-1}) and was referred for parathyroidectomy. Her first surgery involved excision of only one gland. She subsequently underwent a second surgery for residual hyperparathyroidism. Her blood pressure, serum

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Table 1. Clinical and laboratory characteristics of our patients before and after parathyroidectomy

Patient	*Case I; 45-year-old male*	*Case II; 46-year-old female*	*Case III; 30-year-old male*
ESRD	Membranous nephropathy (from age 29)	Congenital anomaly (from age 23)	Congenital disease (from childhood)
Preoperative data			
Average BP before operation (mm Hg)	140/90	180/100–200/100	175/120–210/120 predialysis
Antihypertensive medications	None	Clonidine, atenolol, amlodipine and doxazosin	Clonidine and minoxidil
Last Ca^{++}, PTH preoperative	Ca^{++} 9.74 mg dl^{-1} PTH 2408 pg dl^{-1}	Ca^{++} 9.5 mg dl^{-1} PTH 7000 pg dl^{-1}	Ca^{++} 9.3 mg dl^{-1} PTH 3642 pg ml^{-1}
Immediate postoperative data			
Immediate postoperative Ca^{++}, PTH	Ca^{++} 6.23 mg dl^{-1} PTH 6 pg ml^{-1}	Ca^{++} 7.2 mg dl^{-1} PTH 2000 pg ml^{-1}	Ca^{++} 6.4 mg dl^{-1} PTH 16–30 pg ml^{-1}
Immediate postoperative BP (mm Hg)	60/40	50/30	100/60
Current data			
Dialysis regimen	5 h*3 weekly Filter: Fx-80 Blood flow 300 ml min^{-1}	3.5 h*3 weekly Filter: Fx-80 Blood flow 300 ml min^{-1}	3 h 45 min*3 weekly Filter: Fx-80 Blood flow 300 ml min^{-1}
Current laboratory tests (predialysis; mg dl^{-1})	Creatinine 10.9 BUN 42 Na 139 Phosphor 2.3 Calcium 9.13	Creatinine 9.57 BUN 53 Na 140 Phosphor 4 Calcium 8.77	Creatinine 11.19 BUN 77 Na 139 Phosphor 6.9 Calcium 7.39
Average long-term BP	60/40	80/50–90/60	100/60–105/70
Calcium-phosphor-related medications	Calcium carbonate 600 mg elemental Ca^{++} daily 25(OH)D3 2200 U daily	Calcium carbonate 3600 mg elemental Ca^{++} daily 1 μg 125(OH)2D3 daily	Calcium carbonate 3600 mg elemental Ca^{++} daily. Extra calcium added according to calcium levels
BP raising measures	Midodrine, fludrocortisone, low dialysate temperature (35 °C), high dialysate calcium (3 meq l^{-1}) and sodium (140 meq l^{-1}) Minimal UF 4 ml min^{-1}	Midodrine, low dialysate temperature (35 °C), high dialysate calcium (3 meq l^{-1}) and sodium (140 meq l^{-1}) UF 3 l per HD	Stopped BP meds, low dialysate temperature (35 °C), high dialysate calcium (3 meq l^{-1}) and sodium (140 meq l^{-1}) Minimal UF 2–3 ml min^{-1}

Abbreviations: BP, blood pressure; BUN, blood urea nitrogen; PTH, parathyroid hormone; UF, ultrafiltration.

calcium and PTH values before and after the second operation are shown in Table 1. There were no reports of operative complications or trauma to the carotid body or sinus. Her thyroid function was within normal limits. Echocardiography demonstrated moderate mitral regurgitation, and her hypotension only partially responded to volume expansion. She is currently being treated with midodrine, as well as lifestyle and dialysis modifications. Despite this, she has difficulties completing dialysis sessions because of weakness and presyncope. Orthostatic hypotension was not present—in our clinic, she had a blood pressure of 88/55 mm Hg and heart rate of 69 bpm. On standing, her blood pressure increased to 98/56 mm Hg. Her PWV was 6.4 ± 0.3 m s^{-1}.

CASE III

A 30-year-old male with ESRD since childhood and severe hyperparathyroidism underwent a subtotal parathyroidectomy in 2003. In 2011, he developed 'brown tumors' of his facial bones, with the characteristic histology of osteitis fibrosa cystica, and, therefore, had excision of the remaining parathyroid adenoma (Table 1). He had resistant hypertension with blood pressure values of 170/90–210/120 mm Hg despite a high dose of clonidine and minoxidil, which he used for many years. Immediately after the operation, his blood pressure fell to <100 mm Hg systolic. Echocardiography was unremarkable. At our clinic, his blood pressure was 101/68–95/69 mm Hg with no orthostatic fall. Aortic blood pressure was 95/69 mm Hg. PWV was 7 m s^{-1}, well within the normal range.

These cases demonstrate a strikingly similar occurrence of an acute fall in blood pressure in hemodialysis patients undergoing total or subtotal parathyroidectomy. The low blood pressure, occurring in a previously hypertensive individual, persists for months to years, and is difficult to treat.

The fall in blood pressure coincides with a substantial fall in plasma calcium levels, well-described phenomena after parathyroidectomy, termed as 'hungry bone syndrome'.

We found no similar cases in the medical literature. Shinoda *et al.*[8] describe an ESRD patient who developed acute hypotension and overt heart failure immediately after parathyroidectomy. Unlike our patients, the patient described had a compensated heart failure, which was unmasked by disappearance of PTH, possibly due to decreased influx of calcium into cardiac myocytes, loss of the ionotropic effect of intracellular calcium and development of hypotension with overt heart failure, which responded to calcium infusion.[9,10]

Parathyroidectomy can cause surgical complications, which can be severe, such as postoperative quadriplegia.[11] None of our patients experienced surgical complications, including injury to the carotid body or carotid sinus. None of them had orthostatic hypotension, orthostatic tachycardia or orthostatic hypertension. They had no other evidence of chemoreceptor or baroreceptor dysfunction. Goldsmith *et al.*[4] stressed the discrepancy between

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changes in Ca^{++} and/or PTH that are immediate after the surgical procedure, and changes in blood pressure that occur later and are usually gradual. They argue that there might be a slow process of resetting related to intracellular calcium handling, or gradual removal of excess calcium from the walls of blood vessels after sustained elevation.

Our case series show that changes in calcium and blood pressure can be both abrupt and simultaneous. This drop in calcium level can cause a decrease in intracellular calcium, causing a shift from vasoconstriction to vasodilation. Lack of calcium, together with possible depression of renin, angiotensin and aldosterone might cause an acute resetting of blood vessel tone. For unclear reasons, even if calcium levels normalize, hypotension persists. The hypotension in all three cases was unrelated to volume status, cardiac function or autonomic function. It was related to excessive dilation of arteries, decreased SVR and remodeling of large arteries with low-to-normal arterial stiffness. Larger series and prospective studies are needed to understand hypotension in chronic dialysis patients, and in particular the relation between removal of the parathyroid gland, changes in calcium homeostasis and blood pressure.

What is known about this topic

- Primary, secondary or tertiary hyperparathyroidism are related to secondary hypertension.
- Parathyroidectomy induces a mild and gradual drop in blood pressure. However, this drop is modest in magnitude, and is non-sustained.

What this study adds

- This is the first description of a substantial and long-lasting drop in blood pressure of hemodialysis patients, occurring immediately after an uncomplicated parathyroidectomy operation. This phenomenon ‘converts’ highly resistant hypertension into difficult-to-manage and symptomatic hypotension.
- Hypertension, in many ways, is a mirror image of hypotension. This case series stresses the possible interactions between extracellular and intracellular calcium, parathyroid hormone, vascular smooth muscle cell contractility and blood pressure regulation, in the unusual setting of long-standing renal replacement therapy. A better understanding of the obscure underlying mechanisms might help us manage both resistant hypertension and resistant hypotension.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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